REVIEW FOR FINAL EXAM GRADE 12

A. Multiple choice questions

1. The magnitude of the magnetic field at the center of a circular coil of radius **r**, having **n** turns and carry a current of I can be doubled by:
a. Changing I to 2I, keeping n unchanged
b. Changing n to 2n, keeping I unchanged
c. Changing I to 2I, and n to 2n
d. Changing n to n/2, keeping I unchanged.

2. A straight wire of length 0.20 m moves at a steady speed of 3.0 m/s at right angles to the magnetic field of flux density 0.10 T. The e.m.f induced across ends of wire is
a. 0.5 V **b.** 0.06 V
c. 0.05 V **d.** 0.04 V

3. The magnetic field inside a solenoid
a. The strength of magnetic field depends on the length of the solenoid
b. The strength of magnetic field depends on the number of turns and current in the solenoid
c. The field lines are not uniform at the center of the solenoid
d. The field lines are perpendicular to the axis of the solenoid.

4. Total number of magnetic field lines passing through an area is called
a. Magnetic flux density **b.** magnetic flux
c. electromotive force **d.** voltage

5. The electromotive force can be induced in a circuit by
a. changing magnetic flux density **b.** changing area of circuit
c. changing the angle **d.** all of above

6. The battery of a car is connected to the motor by 50 cm long wires which are 1.0 cm apart. If the current in the wires is 200 A, Find the force between the wires. Is the force attractive or repulsive.
a. 0.4 N, attractive **b.** 0.04 N, attractive
c. 0.4 N, repulsive **d.** 0.04 N, repulsive

7. A current carrying metal wire of diameter 2 mm produces a maximum magnetic field of magnitude 2×10^{-3}. The current in the wire is
a. 10 A **b.** 20 A
c. 40 A **d.** $40\sqrt{2}$ A

8. Three long, straight and parallel wires C, D and G carrying currents are arranged as shown in Fig. 24.49. The force experienced by a 25 cm length of wire C is
a. 0.4 N **b.** 0.44 N
c. 4×10^{-3} N **d.** 4×10^{-4} N

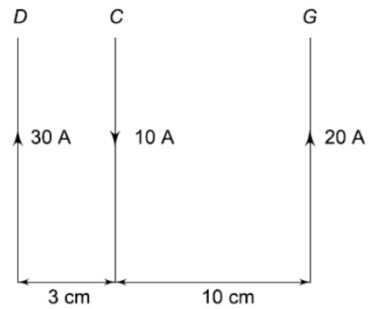


Fig. 24.49

9. A potential difference of 600 V is applied across the plates of a parallel plate capacitor. The separation between the plates is 3 mm. An electron projected vertically, parallel to the plates, with a velocity of 2×10^6 m/s moves undeflected between the plates. What is the magnitude of the magnetic field between the capacitor plates?
a. 0.1 T **b.** 0.2 T
c. 0.3 T **d.** 0.4 T

10. Two straight in finitely long and thin parallel wires are help 0.1 m apart and carry a current of 5 A each in the same direction. The magnitude of the magnetic field at a point 0.1 m from both wires is
a. 10^{-5} T **b.** $\sqrt{2}\times10^{-5}$ T
c. $\sqrt{3}\times10^{-5}$ T **d.** 2×10^{-5} T

11. If the particle emerges from the region of magnetic field with its final velocity at angle of 30° to its initial velocity, the value of l is
a. $\dfrac{2mv_0}{qB}$ **b.** $\dfrac{mv_0}{qB}$
c. $\dfrac{mv_0}{2qB}$ **d.** $\dfrac{\sqrt{3}mv_0}{2qB}$

12. A circuit consist of a 2 μF capacitor connected through an a.c ammeter to an a.c source of voltage given by

$$v(t)=100\sqrt{2}\sin\left(100t+\frac{\pi}{3}\right)$$

Where v is in volt and t in seconds. The reading of the ammeter is
a. 10 mA **b.** 20 mA
c. 30 mA **d.** 40 mA

13. A series RLC circuit with $L = 5$ H, $C = 80$ μF and $R = 40$ Ω is connected to a variable frequency 230 V a.c source.
(a) What is the source frequency which drives the circuit at resonance?
a. 50 Hz **b.** 7.96 Hz
c. 15.92 Hz **d.** 0.796 Hz
(b) What is the impedance of the circuit at resonance?
a. 0 Ω **b.** 40 Ω
c. 100 Ω **d.** 80 Ω
(c) Find peak value of the current at resonance.
a. 5.75 A **b.** 7.00 A
c. 8.13 A **d.** 9.70 A
(d) Find the rms potential differences across L, C and R at resonance.
a. $\begin{cases} V_R = 230 \text{ V} \\ V_L = 1437.5 \text{ V} \\ V_C = 1437.5 \text{ V} \end{cases}$ **b.** $\begin{cases} V_R = 1437.5 \text{ V} \\ V_L = 230 \text{ V} \\ V_C = 1437.5 \text{ V} \end{cases}$

c. $\begin{cases} V_R = 230\ \text{V} \\ V_L = 115\ \text{V} \\ V_C = 115\ \text{V} \end{cases}$ **d.** $\begin{cases} V_R = 230\ \text{V} \\ V_L = 0\ \text{V} \\ V_C = 0\ \text{V} \end{cases}$

(e) What is the total potential difference across the combination of L and C at resonance?

a. 1437.5 V **b.** 230 V
c. 2875 V **d.** 0 V

(f) Find the maximum power transferred to the circuit from the source in one complete cycle.

a. 1330 W **b.** 1322.5 W
c. 1400 W **d.** 1200 W

14. A 2 MeV proton is moving perpendicular to uniform magnetic field of 2.5 T. The force on the proton is

a. 2.5×10^{-10} N **b.** 8×10^{-11} N
c. 2.5×10^{-11} N **d.** 8×10^{-12} N

15. A wire of length 0.50 m carries a current of 0.10 A in the positive x-direction, parallel to the ground. If the wire has a weight of 1.0×10^{-2} N, what is the minimum magnitude magnetic field that exerts a magnetic force on the wire equal to the wire's weight?

a. 0.20 T **b.** 0.30 T
c. 0.40 T **d.** 0.50 T
e. 0.60 T

16. A proton enters a constant magnetic field of magnitude 0.050 T and traverses a semicircle of radius 1.0 mm before leaving the field. What is the proton's speed?

a. 1.6×10^3 m/s **b.** 2.3×10^3 m/s
c. 2.8×10^4 m/s **d.** 1.8×10^2 m/s
e. 4.8×10^3 m/s

17. A 10-turn square coil 0.50 m on a side lies in the xy plane. A uniform magnetic field with magnitude 1.0 T in the negative z-direction changes steadily to 3.0 T in the positive z-direction, a process that takes 2.0 s.
What is the magnitude of the induced emf in the coil?

a. 2.0 V **b.** 5.0 V
c. 3.0 V **d.** 4.0 V
e. 6.0 V

18. The current in a 5.00-H inductor decreases uniformly at a rate of 2.00 A/s. What is the voltage drop across the inductor?

a. 2.50 V **b.** 5.00 V
c. 0.400 V **d.** 10.0 V
e. 7.50 V

19. A generator contains a 100-turn coil that rotates 10.0 times per second. If each turn has an area of 0.100 m^2 and the magnetic field through the coils is 0.050 0 T, what is the maximum emf induced in the coil?

a. 31.4 V **b.** 62.8 V
c. 0.314 V **d.** 6.28 V
e. 3.14 V

20. When a particular inductor is connected to a source of sinusoidally varying voltage with constant amplitude and a frequency of 60.0 Hz, the rms current is 3.0 A. What is the rms current if the source frequency is doubled?

a. 12 A **b.** 6.0 A
c. 4.2 A **d.** 3.0 A **e.** 1.5 A

21. A series RLC circuit contains a resistor of 20 V, a capacitor of 0.75 μF, and an inductor of 120 mH. If a sinusoidally varying rms voltage of 120 V is applied across this combination of elements, what is the rms current in the circuit when operating at its resonance frequency?

a. 2.4 A **b.** 6.0 A
c. 10 A **d.** 17 A **e.** 8.2 A

22. An inductor and a resistor are connected in series across an AC generator, as shown in Figure MCQ21.8. Immediately after the switch is closed, which of the following statements is true?

a. The current is $\frac{V_{\max}}{R}$.
b. The voltage across the inductor is zero.
c. The current in the circuit is zero.
d. The voltage across the resistor is V.
(e) The voltage across the inductor is half its maximum value.

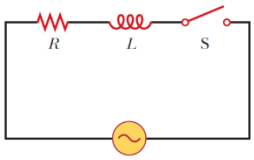


Figure MCQ21.8

B. Critical thinking questions

23. When does the magnetic flux occur?
24. A loop of wire is placed in a uniform magnetic field.
(a) For what orientation of the loop is the magnetic flux a maximum?
(b) For what orientation is the flux zero?
25. When a charged particle enters a uniform magnetic field, under what conditions does it exhibit linear, circular, or helical (spiral) motion?
26. What is the phenomenon of electromagnetic induction?
27. What is the difference between an electric motor and an electric generator?
28. State Lenz's Law to determine the direction of the induced current.
29. What is the inductance of a circuit? On what factors does it depend, and what is its SI unit?
30. In an RL circuit, after closing the switch $t = 1\tau$, $t = 2\tau$, $t = 3\tau$, $t = 4\tau$, $t = 5\tau$ and $t = \infty$, what percentage does the current across the circuit reached of its maximum current?
31. What is electric oscillation? How many types of electric oscillations are there, and what are they?
32. What are the rms current and voltage in AC circuits? What their relationship to the maximum (peak) current and voltage?
33. What is the Fresnel constructed for?

C. Work problems

34. The induced electromotive force induced in a coil of resistan 6.5 Ω has an expression of $e = 6t+5$ (mV).
Find the induced current in the coil at t = 10 s.

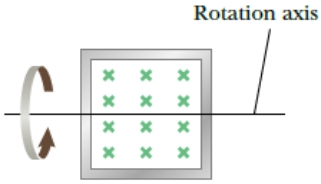
35. A square coil of resistance 2 Ω, 100 turns and side 10cm is placed with its plane making an angle of 300 with a uniform magnetic field of 0.1T. In 0.05s the coil rotates until it plane becomes parallel to the magnetic field. Calculate
a. The induced e.m.f across the two ends of the coils;
b. The current induced in the coil.

36. A particle carries a charge of 5.0×10^{-8} C. The particle is given an initial horizontal velocity of 6.0×10^4 m/s to the right and is perpendicular to the magnetic field of $B = 3.0$ T into page. To keep the particle moving in a horizontal direction, what is the direction and magnitude of
gravitational force acting on particle?

37. The solenoid has 1000 turns of copper wire, the air core is 20cm long, and produces magnetic field of 3.14 mT. Given, $\mu_0 = 4\pi\times10^{-7}\ \mathrm{Tm/A}$. Calculate the electric current passes through the solenoid (in Ampere).

38. A wire carries a current of 10.0 A in a direction that makes an angle of 30.0° with the direction of a magnetic field of strength 0.300 T. Find the magnetic force on a 5.00m length of the wire.

39. A square coil of wire of side 2.80 cm is placed in a uniform magnetic field of magnitude1.25 T directed into the page as in Figure. The coil has 28.0 turns and a resistance of 0.780 V. If the coil is rotated through an angle of 90.0° about the horizontal axis shown in 0.335 s, find
(a) the magnitude of the average emf induced in the coil during this rotation and
(b) the average current induced in the coil during this rotation.



40. Find the magnetic flux of the Earth's magnetic field of magnitude 5 x 10^{-5} T through a square loop 20 cm^2.
a. When the field is perpendicular to the plane of the loop.
b. When the field makes 30 degrees with the normal to the plane of loop.
c. When the field makes 90 degrees with the normal to the plane.

41. A laboratory electromagnet produces a magnetic field of magnitude 1.5 T. A proton (q = +1.610^{-19}C) moves through this field with a speed of v = 6 x 10^6 m/s
a. Find the magnitude of the maximum magnetic force that could be exerted on the proton.
b. What is the magnitude of maximum acceleration of the proton?
c. Would the field exert the same magnetic force on an electron moving through Field with the same speed?

42. A solenoid with a diameter of 0.2 meters has 500 turns per meter. Surrounding the center of this solenoid is a coil consisting of 100 tightly wrapped turns. If the current flowing through the solenoid increases from 0 to 2 A in 1 millisecond, determine the induced emf generated in the coil.

43. The magnetic flux through a coil with a resistance of 6.5 Ω, oriented perpendicular to a uniform magnetic field, varies over time t t (in seconds) according to the equation: $\phi = 3t^2 + 5t + 2\ (\mathrm{mWb})$. Find the induced current in the coil at t =10 seconds.

44. An aircraft with a wingspan of 50 meters is flying eastward at a constant speed of 1080 km/h, maintaining a steady altitude. At this location, the vertical component of the Earth's magnetic field is 2 × 10^{-5} T. Calculate the electromotive force (emf) generated between the tips of the aircraft's wings.

45. An emf of 1 mV is induced in a coil when the current in it changes steadily from 2 A to 4 A in 0.1 seconds. Find the self-inductance of the coil.

46. A solenoid 1.0 m long and 0.05 m in diameter has 700 turns. Another solenoid of 50 turns is tightly wound over the first solenoid. Find the emf induced in the second solenoid when the current in the first solenoid changes from 0 to 5 A in 0.01 seconds.

47. An ideal inductor of inductance of 5 H and a pure resistor of resistance 100 Ω are connected in series to a battery of emf 6 V of negligible internal resistance through a switch. The switch is closed at time t = 0 s.
a. Find the maximum (or steady) value of the current.
b. What is the time constant τ of the circuit.
c. How long does it take for the current to rise to 50% of its maximum value?
d. Find the potential difference across the inductor at t = 0.1 s. ***Given:*** $e^{-2} = 0.135$

48. An inductor of inductance 10 H and a resistor of resistance 16 Ω are connected to a 12 V dc source.
a. Find the final steady current.
b. How much energy is consumed to attains this steady current?
c. What is the power dissipated in the resistor at this current?

49. An inductor of inductance 100 mH and a resistor of Resistance 10 Ω are connected in series to a 2 V battery. After some time is now short circuited. Calculate the time required for the current fall to haft the steady value.

50. An alternating voltage V (in volt) varies with time t (in second) as $v(t) = 100\sin(50\pi t)$. Find the peak value, rms value and frequency of the alternating current.

51. A 100 Ω electric iron is connected to a 200 V, 50 Hz a.c supply. Find (a) rms value of voltage, (b) peak value of voltage (c) rms value of current and (d) peak value of current.

52. A coil of inductance 0.5 H and a resistor of resistance 100 Ω are connected in series to a 240 V, 50 Hz supply.
a. Find the maximum current in the circuit.
b. Find the time lag between maximum voltage and current.